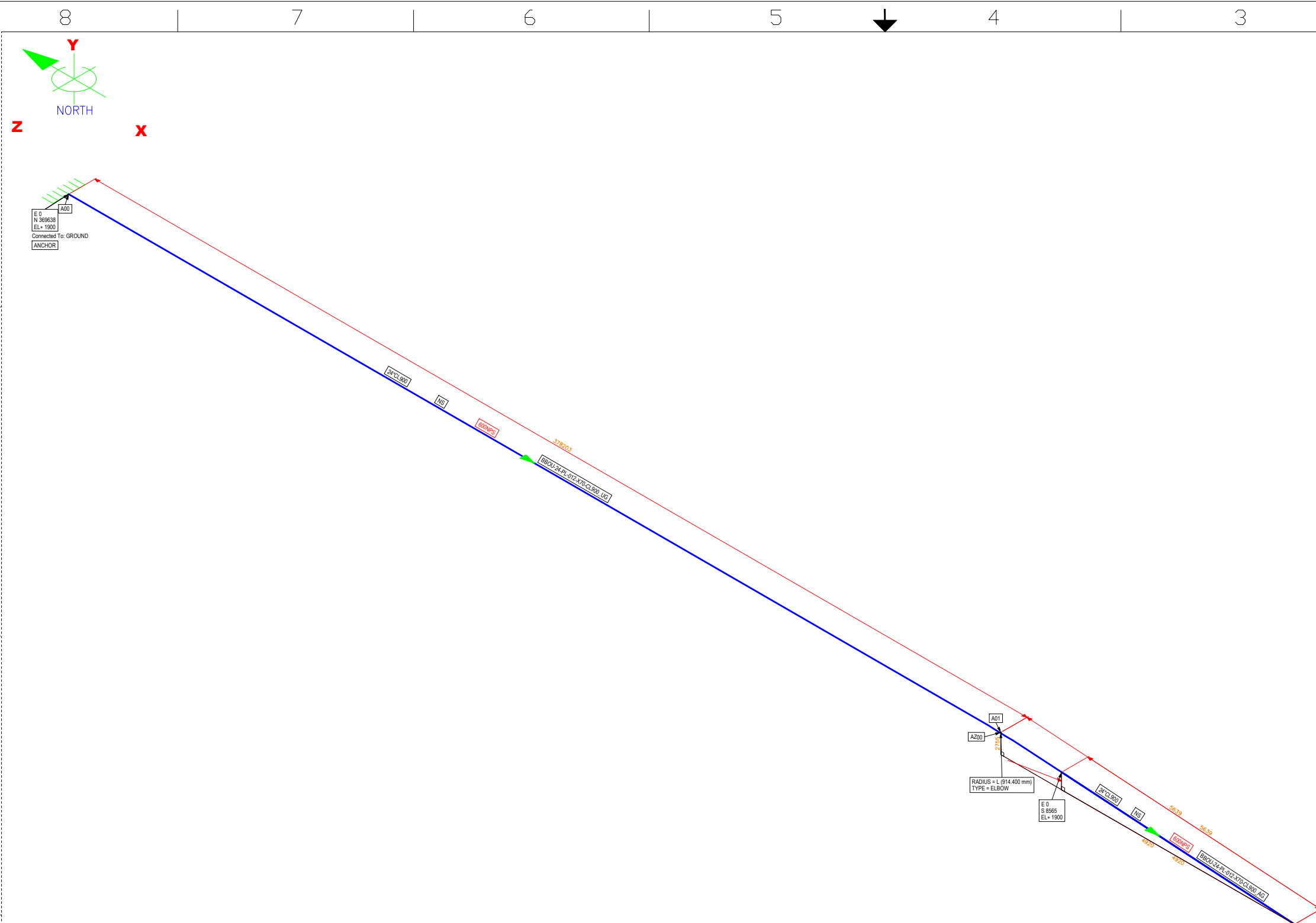




```

MODEL INFO                                     AutoPIPE Ver: 12.06.00.48
System: BBDU _ASME B31.8
Piping Code: ASME B31.8 - 2016
Model Rev: 9                               Ambient: 24.0 (deg C)

-----
PIPE DATA (mm)
-----
Pipe ID      Material      Nom      Sch      D.D.      W.Th      Corr.      Mill      Insul      SG

24"CL900    SLX-X70              600.00    NS 609.60 15.880  3.000  1.985              0.700

Gap1: Vstop, Guide-V=down, Lstop, Incl, Tielink=back, Guide-H=Left
Gap2: Vstop, Guide-V=Up, Lstop, Incl, Tielink=fwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

For Rotational Supports: Loads are moments (ft-lb) and
movements are rotations (deg)

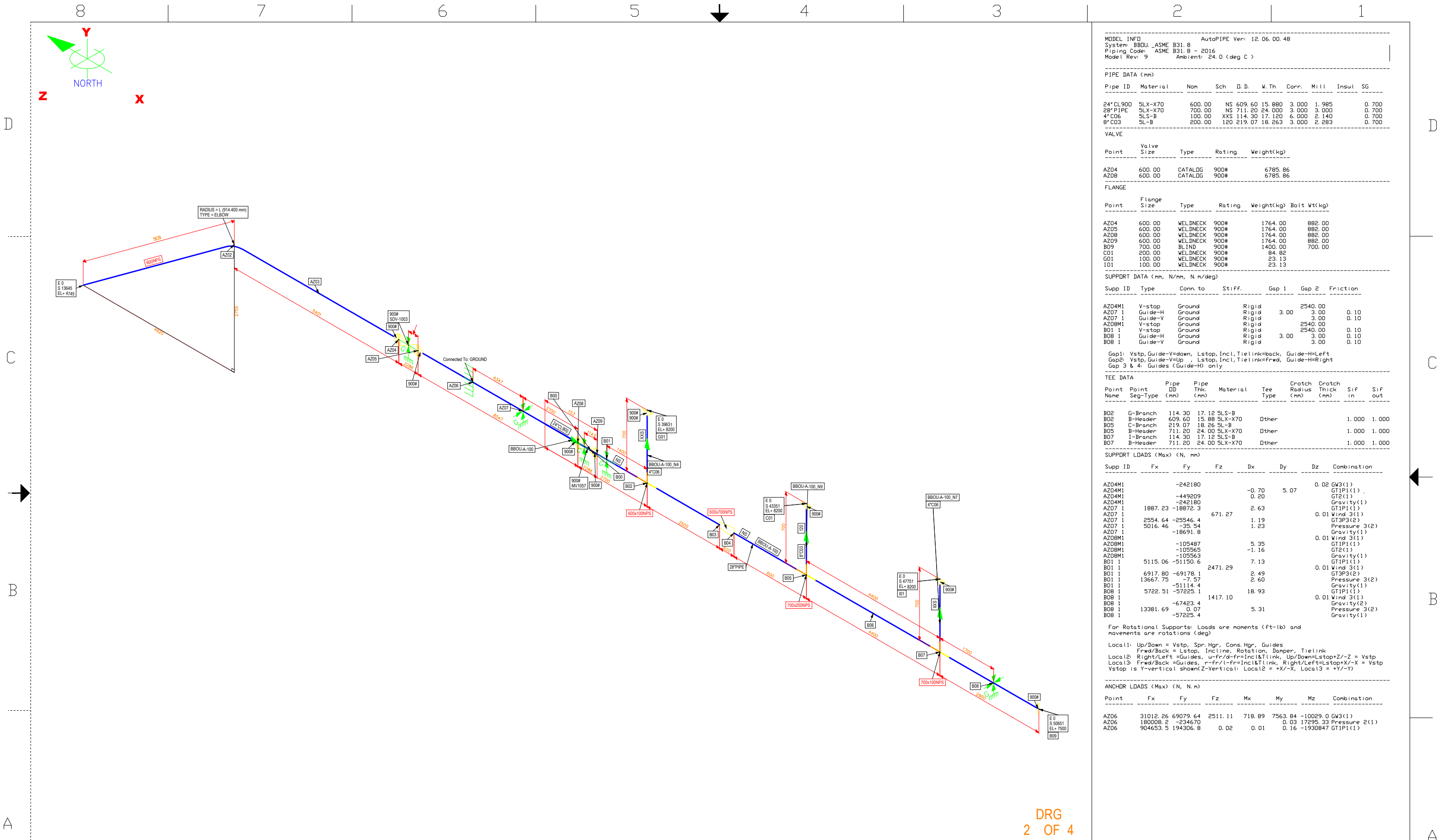
Local1: Up/Down = Vstop, Spr,Hgr, Cons,Hgr, Guides
        Fwd/Back = Vstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left = Guides, up/fr/l=fr>Incl&Tlink, Up/Down=Lstop+Z = Vstop
Local3: Fwd/Back = Guides, -fr/r/l=fr>Incl&Tlink, Right/Left=Lstop+X/-X = Vstop
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

-----
ANCHOR LOADS (Max) (N, N.n)
-----
Point      Fx      Fy      Fz      Mx      My      Mz      Combination

A00              -0.12   -9015.25              23.82              -11845.7 GW3(1)
A00              -0.15   -12398.8              23.82              -16239.0 Gravity(2)
A00      -3653490   -9015.25              23.82              -11845.7 GT1P1(1)

```

[illegible]



MODEL INFO				AutoPIPE Ver: 12.06.00.48					
System: BBOU_ASM B31.8									
Piping Code: ASME B31.8 - 2016									
Model Rev: 9				Ambient: 24.0 (deg C)					
PIPE DATA (mm)									
Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
24"CL900	SLX-X70	600.00	NS	609.60	15.880	3.000	1.985		0.700
28"PIPE	SLX-X70	700.00	NS	711.20	24.000	3.000	3.000		0.700
4"CO6	SLX-B	100.00	XXS	114.30	17.120	6.000	2.140		0.700
8"CO3	SL-B	200.00	120	219.07	18.263	3.000	2.283		0.700
VALVE									
Point	Valve Size	Type	Rating	Weight(kg)					
AZ04	600.00	CATALDG	900#	6785.86					
AZ08	600.00	CATALDG	900#	6785.86					
FLANGE									
Point	Flange Size	Type	Rating	Weight(kg)		Bolt Wt(kg)			
AZ04	600.00	WELDNECK	900#	1764.00		882.00			
AZ05	600.00	WELDNECK	900#	1764.00		882.00			
AZ08	600.00	WELDNECK	900#	1764.00		882.00			
AZ09	600.00	WELDNECK	900#	1764.00		882.00			
B09	700.00	BLIND	900#	1400.00		700.00			
C01	200.00	WELDNECK	900#	84.82					
I01	100.00	WELDNECK	900#	23.13					
SUPPORT DATA (mm, N/mm, N.m/deg)									
Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction			
AZ04M1	V-stop	Ground	Rigid		2540.00	0.10			
AZ07 1	Guide-H	Ground	Rigid	3.00		0.10			
AZ07 1	Guide-V	Ground	Rigid		3.00	0.10			
AZ08M1	V-stop	Ground	Rigid		2540.00	0.10			
B01 1	V-stop	Ground	Rigid		2540.00	0.10			
B08 1	Guide-H	Ground	Rigid	3.00		0.10			
B08 1	Guide-V	Ground	Rigid		3.00	0.10			
Gap1: Vstop, Guide-V=down, Lstop, Incl, Tielink=back, Guide-H=Left Gap2: Vstop, Guide-V=Up, Lstop, Incl, Tielink=Frwd, Guide-H=Right Gap 3 & 4: Guides (Guide-H) only									
TEE DATA									
Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk. (mm)	Material	Tee Type	Crotch Radius (mm)	Crotch Thick (mm)	Sif in	Sif out
B02	G-Branch	114.30	17.12	SLX-B					
B02	B-Header	609.60	15.88	SLX-X70	Other			1.000	1.000
B05	C-Branch	219.07	18.26	SL-B					
B05	B-Header	711.20	24.00	SLX-X70	Other			1.000	1.000
B07	I-Branch	114.30	17.12	SLX-B					
B07	B-Header	711.20	24.00	SLX-X70	Other			1.000	1.000
SUPPORT LOADS (Max) (N, mm)									
Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination		
AZ04M1		-242180				0.02	GW3(1)		
AZ04M1			-0.70		5.07		GT1P1(1)		
AZ04M1		-449209	0.20				GT2(1)		
AZ04M1		-242180					Gravity(1)		
AZ07 1	1887.23	-18872.3		2.63			GT1P1(1)		
AZ07 1			671.27				GT3P3(2)		
AZ07 1	2554.64	-25546.4		1.19			Pressure 3(2)		
AZ07 1	5016.46	-35.54		1.23			Gravity(1)		
AZ08M1		-18691.8					GT1P1(1)		
AZ08M1		-105487	5.35				Wind 3(1)		
AZ08M1		-105565	-1.16				GT1P1(1)		
AZ08M1		-105563					GT2(1)		
B01 1	5115.06	-51150.6		7.13			Gravity(1)		
B01 1			2471.29				GT1P1(1)		
B01 1	6917.80	-69178.1		2.49			Wind 3(1)		
B01 1	13667.75	-7.57		2.60			GT3P3(2)		
B01 1		-51114.4					Pressure 3(2)		
B08 1	5722.51	-57225.1		18.93			Gravity(1)		
B08 1			1417.10				GT1P1(1)		
B08 1		-67423.4					Wind 3(1)		
B08 1		0.07					Gravity(2)		
B08 1	13381.69			5.31			Pressure 3(2)		
B08 1		-57225.4					Gravity(1)		
For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)									
Local1: Up/Down = Vstop, Spr.Hgr, Cons.Hgr, Guides Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink Local2: Right/Left = Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstop Local3: Frwd/Back = Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstop Vstop is Y-vertical shown(Z=Vertical) Local2 = +X/-X, Local3 = +Y/-Y									
ANCHOR LOADS (Max) (N, N.m)									
Point	Fx	Fy	Fz	Mx	My	Mz	Combination		
AZ06	31012.26	69079.64	2511.11	718.89	7563.84	-10029.0	GW3(1)		
AZ06	180008.2	-234670			0.03	17295.33	Pressure 2(1)		
AZ06	904653.5	194306.8	0.02	0.01	0.16	-1930847	GT1P1(1)		

REV.

DATE

REVISION

BY

CHKD

APP

G-Gap

ANCHOR

LINESTOP

V-STOP

ROTATION

GUIDE

INCLINED

SPRING

CONSTANT

DAMPER

Line Direction For Support Gaps

PROJECT NO: AKABEL

ANALYZED BY: T. D

CHECKED BY: E. N

APPROVED BY:

AREA:

LINE NO.:

CLASS:

FLOW DIAGRAM:

DATE:

DATE:

DATE:

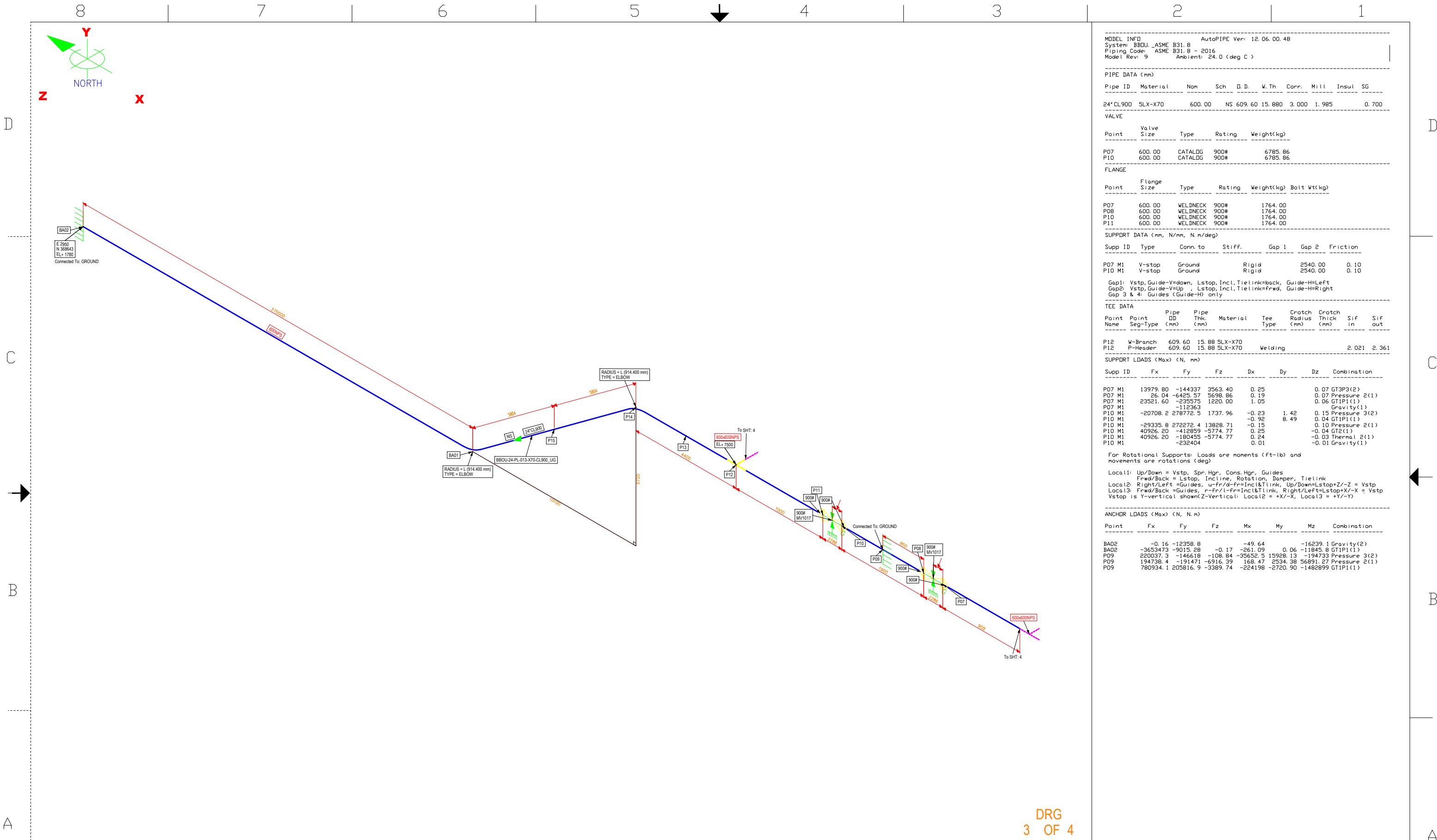
DATE:

DRAWING No.

REV.

DRG SIZE

D



MODEL INFO									
System: BBOU_ASME B31.8 AutoPIPE Ver: 12.06.00.48									
Piping Code: ASME B31.8 - 2016									
Model Rev: 9 Ambient: 24.0 (deg C)									
PIPE DATA (mm)									
Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
24"CL900	SLX-X70	600.00	NS	609.60	15.880	3.000	1.985		0.700
VALVE									
Point	Valve Size	Type	Rating	Weight(kg)					
P07	600.00	CATALOG	900#	6785.86					
P10	600.00	CATALOG	900#	6785.86					
FLANGE									
Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)				
P07	600.00	WELDNECK	900#	1764.00					
P08	600.00	WELDNECK	900#	1764.00					
P10	600.00	WELDNECK	900#	1764.00					
P11	600.00	WELDNECK	900#	1764.00					
SUPPORT DATA (mm, N/mm, N.m/deg)									
Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction			
P07 M1	V-stop	Ground	Rigid		2540.00	0.10			
P10 M1	V-stop	Ground	Rigid		2540.00	0.10			
Gap1: Vstop,Guide=Vdown, Lstop,Incl,Tielink=back, Guide=H=Left									
Gap2: Vstop,Guide=Vup, Lstop,Incl,Tielink=frwd, Guide=H=Right									
Gap 3 & 4: Guides (Guide-H) only									
TEE DATA									
Point Name	Point Seg-Type	Pipe DD (mm)	Pipe Thk. (mm)	Material	Tee Type	Crotch Radius (mm)	Crotch Thick (mm)	Sif in	Sif out
P12	W-Branch	609.60	15.88	SLX-X70					
P12	P-Header	609.60	15.88	SLX-X70	Welding			2.021	2.361
SUPPORT LOADS (Max) (N, mm)									
Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination		
P07 M1	13979.80	-144337	3563.40	0.25		0.07	GT3P3(2)		
P07 M1	26.04	-6425.57	5698.86	0.19		0.07	Pressure 2(1)		
P07 M1	23521.60	-235575	1220.00	1.05		0.06	GT1P1(1)		
P07 M1		-112363					Gravity(1)		
P10 M1	-20708.2	278772.5	1737.96	-0.23	1.42	0.15	Pressure 3(2)		
P10 M1				-0.92	8.49	0.04	GT1P1(1)		
P10 M1	-29335.8	272272.4	13828.71	-0.15		0.10	Pressure 2(1)		
P10 M1	40926.20	-412859	-5774.77	0.25		-0.04	GT2(1)		
P10 M1	40926.20	-180455	-5774.77	0.24		-0.03	Thermal 2(1)		
P10 M1		-232404		0.01		-0.01	Gravity(1)		
For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)									
Local1: Up/Down = Vstop, Spr.Hgr, Cons.Hgr, Guides									
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink									
Local2: Right/Left = Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstop									
Local3: Frwd/Back = Guides, u-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstop									
Vstop is Y-vertical shown(Z=Vertical) Local2 = +X/-X, Local3 = +Y/-Y									
ANCHOR LOADS (Max) (N, N.m)									
Point	Fx	Fy	Fz	Mx	My	Mz	Combination		
BA02	-0.16	-12358.8		-49.64		-16239.1	Gravity(2)		
BA02	-3653473	-9015.28	-0.17	-261.09	0.06	-11845.8	GT1P1(1)		
P09	220037.3	-146618	-108.84	-35652.5	15928.13	-194733	Pressure 3(2)		
P09	194738.4	-191471	-6916.39	168.47	2534.38	56891.27	Pressure 2(1)		
P09	780934.1	205816.9	-3389.74	-224198	-2720.90	-1482899	GT1P1(1)		

REV.

DATE

REVISION

BY

CHKD

APP

G-Gap

ANCHOR

LINESTOP

V-STOP

ROTATION

GUIDE

INCLINED

SPRING

CONSTANT

DAMPER

Line Direction For Support Gaps

PROJECT NO: AKABEL

ANALYZED BY: T. B

CHECKED BY: E. N

APPROVED BY:

AREA:

LINE NO.:

CLASS:

FLOW DIAGRAM:

DATE:

DATE:

DATE:

DATE:

DRAWING No.

DRG. SIZE

D

REV.

8

7

6

5

4

3

2

1

